Open Concerns

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Every passage the final report marks as queried, and the status of each of its eight result slots, collected into one list. 53 items, from the final report.

A flag marks a result, number or condition that is *measured but not yet trustworthy as evidence* — an unexplained mechanism, a confound, an unfair comparison, or a figure whose provenance does not support the use it is being put to. It deliberately does not mark a bad result: a large error that is correctly measured and understood is a finding, not a concern.

The report's source still carries every one of these passages beside the claim it qualifies, but the report no longer renders them: queries are typeset as plain text or dropped, and the status boxes are dropped. This document is where they are read. References such as `[sec:example]` point into the report.

Front matter

1. [To be written by the author. The template reserves this page for thanks to the project advisor, the examination committee, Gensurv Robotics, the Institute of Field Robotics, and any funding source. Nothing has been drafted here on the author's behalf.]

Chapter 3 — Methodology

The Pipelines as Implemented: ORB-SLAM3 and semantic feature rejection

1. *table row:* `loopClosing` | 1 | comment in the same file says “set to 0 deliberately”

The Pipelines as Implemented: Navigation and relocalisation integration: the `vslam_navigation` package

2. *table row:* Forward collision check | off | on | asymmetric; see below

3. *table row:* Forward collision check | off | on | asymmetric; see below

4. The real rig keeps it on, untested, because its prior map is the less trustworthy of the two and no run yet exists to show which failure matters more there.

5. The published speed is also capped in the robot’s own command units, which are not metres per second and whose scale is known only at a single operating point (the speed scale is provisional; [sec:real-platform]).

6. Each run of the sweep writes a manifest beside its data recording the settings it used, the full launch command, the git commit and the wall duration, and `script/aisle_sweep.py -analyse` reads them into a single table, so a result cannot be attributed to the wrong configuration. An earlier revision of this chapter stated that the sweep had no scored summary and that its run directories recorded the estimator configuration but not the relocalisation parameters. That is no longer true and the statement is withdrawn. Its one carried conclusion — that run-to-run spread over identical settings is as wide as most differences between settings — is used in [sec:exp4-sim] to refuse comparisons.

Simulation Platform: Computing and software environment

7. *table row:* OpenCV (Python) | 4.11.0 | differs from the 4.5.4 the estimators link

Simulation Platform: Simulated sensor rig and platform

8. *table row:* `gyr_n` | 0.002 | 0.00030–0.00036 | $\mathrm{rad\,s^{-1}/\sqrt{Hz}}$

9. [Queried] The configured gyroscope noise density is 2×10^{-3} whereas the stationary windows of the four recordings give 3.0×10^{-4}–3.6×10^{-4}, so the configured value is five to seven times too large. The configured accelerometer density of 0.02 sits inside the measured spread, but that spread is itself wide, 0.015–0.024 across four recordings of the same simulated sensor. Both estimators therefore run on an inertial noise model that does not describe this IMU closely. The bias random walks cannot be checked at all: the longest stationary interval in any recording is 2.16 s and an Allan-variance estimate needs minutes. This discrepancy is not incidental: by the design decision of [sec:proprio-method] the EKF baseline is run on these same configured values, so the comparison between the proprioceptive and visual families is made at a shared handicap rather than at each family’s best.

10. [Queried] The accelerometer characterisation is not stable across recordings of the same simulated sensor. The x bias ranges from 0.013 to 0.079 $\mathrm{m\,s^{-2}}$ and the x scale from 1.017 to 1.095 over the four recordings, while the gyroscope z scale agrees to within 0.0007 over the same four. Because the stationary window is between 1.14 and 2.16 s in every case, part of that spread is estimator variance rather than a property of the sensor, and the accelerometer terms should not be quoted as calibrated constants.

Simulation Platform: Simulation fidelity and its limits

11. *table row:* Accelerometer, per axis ($\mathrm{m\,s^{-2}}$) | 0.0566 | 4.002×10^{-3} | 1.5×10^{-2}–2.4×10^{-2} | 2.00×10^{-3}

12. *table row:* `slip1`, `slip2` | 0 on all grounds | an ODE feature; `dartsim` does not implement it, so inert at any value

13. [Threat to validity] This is a bias with a known direction. The EKF baseline of [sec:results-proprio] is evaluated on a floor where the wheels effectively cannot slip and with an encoder of unlimited resolution, which is dead reckoning's best case. Both idealisations favour the proprioceptive estimators over the visual ones, and neither has been relaxed. Any conclusion that a wheel-plus-IMU filter matches or beats the visual pipelines should be read as holding *under these conditions*, not as a general result. `script/make_slippery_floor.py` generates ground models at a chosen friction coefficient so that traction can be made an experimental factor, and `script/encoder_sim.py` imposes a realistic quantisation floor. Neither has been used in any result reported here.

14. *table row:* `gyr_w` | 0.0001 | 1.735×10^{-6} | 58× | gyroscope bias random walk

15. [Retired evidence for the inflated setting] The inflated configuration is retained for every result in this report, on the strength of an experiment that is no longer part of the reported campaign: in September 2026 both estimators were re-run on configurations carrying the true injected densities, and although accuracy improved wherever a run completed, three of ten runs diverged outright where the inflated configuration diverged in none. That evidence was recorded on recordings retired on 6 November 2026, so it is reported here as the reason for a setting rather than as a result, and the comparison has not been repeated on the current recordings.

Simulation Platform: Realistic logistics layouts

16. The generator reproduces the U world exactly but no longer reproduces the I world: it would now emit eight dock-line items instead of thirteen and two more clutter instances. The committed world file is the recorded environment and is the source of these counts.

17. [Visual verification] The generator-side checks are complete, and the first Gazebo session on 6 October 2026 found and fixed one defect the checks could not see (a jack setting its load down and then stopping on the neighbouring cell, because the drop command carried one angle for both the parked load and the approach heading). Three things remain to be confirmed by eye in the simulator rather than by the checker: that the loads ride on the fork side of the jack, the shading of the generated door, trailer and yard meshes, and the open/closed state of each door. Until then the worlds are **implemented** in geometry and **partial** in visual verification.

18. [Provenance] None of the six recordings carries a run manifest of its own: the world each was recorded in is known from the manifests of the replays made on it, not from the recording. Those manifests agree, and the object inventory each bag reports is consistent with the layout they name, but a recording whose world is attested only by later use is weaker evidence than one that records its own world. A manifest should be written at record time, not at replay time.

Real Platform: AMR mobile robot testbed

19. The drivetrain is a rear brushless motor (Flipsky 6384) through a BLDC drive with a Dynamixel RX-64 steering servo, a GigaDevice GD32407V microcontroller on a CAN link to the NRU-51V, an LM2596 regulator from a lithium-polymer pack, and a Wi-Fi router for tele-operation from a smartphone. The faculty template describes the base as a differential-drive robot; the hardware diagram (a single traction motor with a steering servo) and the simulated model of [fig:ackermann-robot] are Ackermann. The description here follows the hardware and should be confirmed by the author.

Real Platform: Sensors and computing unit

20. [Attribution of reproduced vendor material] [fig:camera-dimensions, fig:nru51v-photo, fig:nru5 and [tab:camera-spec, tab:hardware-spec] reproduce manufacturer drawings, photographs and specifications for the AC-IMX390-H190 camera and the Neousys NRU-51V. The faculty manual requires every reproduced table and figure to carry a citation number and a (Source : ...) line beneath it, and none of them does: the datasheet titles, publishers and years were not recorded when the material was taken. Each needs its vendor document identified and added to the reference list before submission. No measurement in this report is taken from these items; they are descriptive.

Real Platform: Real sensor rig and calibration limits

21. *table row:* Second-camera intrinsics | reuse of the first camera | approximation

22. *table row:* Marker-to-IMU lever arm | not measured | placeholder; bounds every real accuracy figure

The Four Experiments: Experiment 1: Estimator comparison against a non-visual baseline

23. *table row:* accelerometer x scale | 1.0321 | 1.0954 | 1.0172 | 1.0339 | –

24. *table row:* accelerometer x bias | 0.038 | 0.013 | 0.013 | 0.079 | $\mathrm{m\,s^{-2}}$

25. *table row:* yaw-rate scale, bicycle | 1.0245 | 0.9321 | 1.0070 | 1.0171 | –

26. σ_b, the bias random walk, is assumed rather than measured: the longest stationary window in any recording is 2.16 s and an Allan-variance estimate needs minutes.

27. [One tuned constant] The real baseline has a parameter fitted against the same tracker it is scored on, albeit on a different recording. A measured scale, from the wheel circumference and the gear ratio or from a tape-measured straight run, would remove the objection; until then the real-domain comparison between this baseline and the visual pipelines is weaker than its simulated counterpart.

Chapter 4 — Experiments and Results

Experiment 1: Estimator Comparison Against a Non-Visual Baseline: Simulation

1. **Status: completed.** Four dataset_allsensor_realistic recordings, two layouts each recorded without and with warehouse traffic. Two families of estimator are scored over the full common interval of each: the non-visual ones — wheel odometry, IMU dead reckoning, an EKF at the visual systems' own inertial noise configuration and a measured-noise EKF as diagnostic — and the visual ones, VINS-Fusion once per recording and ORB-SLAM3 with a CPU or GPU feature extractor and its loop closer active or disabled. Traffic is noted as a condition, not tested as a factor. The latency and resource measurement is taken from eighteen repeated runs on the two mapping recordings, three repeats of each of six configurations, and those repeats are the only within-configuration spread the campaign measures.

2. VINS-Fusion carries a smaller signature of the same kind, pitch and roll RMS of 1.05 and 1.33° against a yaw RMS of 0.24°; whether that is a residual gravity error or the rigid fit absorbing a tilt has not been separated, and it is reported as an observation.

3. [Confound: run order] The six GPU runs were made consecutively from 21:04 to 22:04 and the six CPU runs from 23:27 to 00:18, so the I-recording GPU repeats were the fourth to sixth GPU runs of a session on a laptop that throttles at 90 °C; the front-end record notes that timing comparisons need a cool-start, interleaved

order. The six VINS-Fusion runs are a separate session of 7 November and are not a control for that state. The U-recording figures are the ones to quote until the I recording is repeated interleaved.

4. [Resolved: `realistic_002/loopclosing_gpu`] Earlier on 6 November this run was reported as closure-free from its `loop_closing.csv` and queried. Its manifest shows the real cause: it was launched on the `realistic_003` bag with `realistic_002`'s output path, so it was scored against the wrong truth (4.94 m, 48.9°, 139 % of the recording's frames processed). It is set aside, not deleted; the configuration was re-run on the right bag the same evening and every table here carries that run.

5. [Open: the false closure is diagnosed from its correction, not from the images] The 14 m correction and the one-aisle keyframe separation identify the `realistic_001` closure as false, and both front ends agree; the matched image pair has not been inspected, and the two April-style guards that would catch it (a covisibility-consistency window, or a maximum correction against the inertial drift bound) are untested. Until one is, the finding is a diagnosis, not a fix.

Experiment 1: Estimator Comparison Against a Non-Visual Baseline: Real robot

6. **Status: partial.** Four real recordings with an independent planar reference (`dataset_realv6_000`, `dataset_realv7_000` to `_002`). A non-visual baseline exists: wheel+gyro dead reckoning with the feedback speed scaled by a factor fitted once against the tracker on `realv7_000` and held on the other three. The visual arm is ORB-SLAM3 in four configurations on each recording, sixteen runs, all made with the fork's patch-15 initialisation switches and scored with no repair applied; they give 0.404–0.480 m on `realv6_000`, 0.251–0.268 on `realv7_000`, 0.442–1.278 on `_001` and 0.172–0.198 on `_002`. Across the held recordings the two families are in the same accuracy class, the baseline leading on one, tying on one and losing the third. The EKF of the simulation slot cannot be formed, because the rig publishes no steering measurement; VINS-Fusion has no valid run on any recording; and every value is provisional until the marker-to-IMU lever arm is measured.

7. [Open: the realv6 scale] On `realv6_000` the path-length ratio gives $k = 0.43$ where the three `realv7` bags give 0.32 to 0.33, yet the held $k = 0.325$ scores a path ratio of 1.04 on the matched samples. The export is at 1.7 Hz and fails the validity rules, and the recording predates `realv7` by three days; whether the firmware unit changed or the sparse export misstates the path length is not decided, and the `realv6` row is a diagnostic.

8. [Open: what the improvement is owed to] Two things differ between the two sets of runs, not one. Patch 15 is the first; the second is that `IMU.NoiseGyro` and `IMU.NoiseAcc` were inflated tenfold on 3 November, and no artefact records which values that afternoon's runs used, because `run_metadata.csv` stores the configuration's path and not its contents. The loss of the tilt is patch 15's: the upstream rule tests keyframe spacing below 2 cm within 10 s and cannot be influenced by an inertial noise density. Any part of the accuracy difference is not separable on the present evidence. One control run, the same bag at today's noise with both switches returned to upstream, would separate them and has not been made.

9. [Provenance: the 3 November estimates are gone] The re-runs overwrote `output/output_orb/real/dataset_realv7_00*/` in place, so the pre-patch-15 estimates of all three recordings no longer exist and neither `[tab:realv7-tilt]` nor `[fig:realv7-tilt]` can be regenerated; the frozen copies beside the generator are the only record of the behaviour this section reports as corrected. A run directory should be written under a name that carries its configuration, so that a re-run cannot overwrite the evidence for the behaviour it replaces.

10. [Sparse reference] The tracker ran at 1.7 Hz rather than its nominal 7 Hz on `realv6`, and the validity rules were written for the faster export. Either the export has to be re-captured at full rate or a rule for sparse references has to be written before any value from this recording is quoted outside a diagnostic.

Experiment 2: Offline YOLO Detector Performance: Simulation

11. **Status: completed.** Four realistic-warehouse recordings, 12 698 left-camera frames, scored on 7 November 2026 against projected simulator ground truth with the four-class checkpoint (`weight/runs_1/detect/train/weights/best.pt`: bin, box, bucket, forklift), whose fourth class fires on the pallet jacks that are the moving objects of these recordings. Pooled precision 0.747; recall 0.502 over the truth boxes the exemptions keep and 0.288 over every truth box; moving-object recall 0.288. The walking people are simulator actors with no pose and no class, so they are outside the score on both sides.

Experiment 2: Offline YOLO Detector Performance: Real robot

12. **Status: planned.** Real imagery exists; no real ground-truth boxes do.

Experiment 3: Dense Map Reconstruction and Map-Quality Evaluation: Simulation

13. **Status: partial.** Five databases are scored for geometry and drivability: the tiny warehouse, which passes, and four realistic-warehouse maps — two mapping recordings, one per layout, each built closure-free under two back-end configurations, A and B — all of which fail the quality criterion fixed before scoring. The tiny map's independent drivability reference remains lost.

14. [Queried: reproducibility of the drivability half] The geometry result is fully reproducible: both `map/map_tiny.db` and the world file are present. The drivability result is not. Its reference run, `dataset_static_tiny_repeat`, is absent from `dataset/` and from the removable drive, and the only surviving recording of this warehouse is the one that built the map, which would answer a weaker question ([sec:map-drivability]). The database additionally stores an empty parameter set, so the configuration that produced it is not recoverable from the file, and the RTAB-Map engineering record's claim that the node stamps match no dataset is contradicted by the pose agreement reported above. The drivability figures should therefore be read as a dated measurement rather than a standing result, and should not be carried into the major report until an independent route has been recorded and retained.

15. The failure was attributed there to the front end under repeated structure, not to the mapping back end. No database, run directory or figure for this attempt is registered in the repository at the cutoff, so the result is reported on the strength of the progress record and cannot be audited. A database identified from its contents as this attempt (18 nodes, 37 m along one aisle, poses leaving the floor, placeable in no large world) was found among the unregistered databases on 22 October 2026 and deleted with them on 6 November, unscored. The attempt must be repeated with its artifacts retained before it is carried into any conclusion.

16. [Queried: front end of the realistic builds] RTAB-Map took its pose from TF, so the odometry came from whichever front end was running; none of the four databases records which, and the build Bs were made during the latency sessions of Experiment 3, whose load on the front end is not recorded either. The precision loss cannot be placed on the front end against the back end, nor the November drop on parameters against session, until the build manifest names both.

Experiment 3: Dense Map Reconstruction and Map-Quality Evaluation: Real robot

17. **Status: planned.** No RTAB-Map database has been built from real imagery, and no reference geometry of the site exists to score one against.

Experiment 4: Autonomous Navigation on the Reconstructed Map: Simulation

18. **Status: partial.** Seven live runs on 12 November 2026, in three of the four worlds, measure the localisation against ground truth over hand-driven routes of 21–153 m. No goal was commanded in any of them, so the goal-reaching measures of the reporting rules below are still unused and that half of the experiment remains **planned**. The protocol, goal set and success criteria are now fixed in code (`script/exp4_protocol.yaml`, generated by `script/exp4_goals.py`), which is what the previous revision of this document recorded as missing.

19. The ground truth reports the model origin and every estimate reports the camera/IMU body, so the 0.338 m lever arm is applied to the ground truth before the fit and before any comparison, and never to recorded data. An earlier revision fitted against the uncorrected ground truth and reported the resulting 0.337 m residual as confirmation that the fit was right, on the grounds that it equalled the lever arm. That was the bias, not the evidence: a rigid transform cannot absorb an offset that rotates with the body, so it absorbed part of it as translation and left the rest as residual.

20. **Reporting rules, goal-reaching.** Fixed before any run exists, so that the result cannot be chosen after the fact, and left unchanged here. A goal counts as reached when the base frame is within 0.25 m and 10° of it and the planner reports success.

21. **Reporting rules, pose tracking.** Defined after these data existed, and reported as the weaker claim it is. Error is the planar distance between the published `map` → `base_link` transform, sampled at 2 Hz, and the lever-arm-corrected ground truth interpolated to the same instants.

Experiment 4: Autonomous Navigation on the Reconstructed Map: Real robot

22. **Status: planned.** No real-robot navigation run has been made, and the real rig has no map to navigate on: the dense-map slot of `[sec:exp4-real]` is itself planned, because no RTAB-Map database has been built from real imagery and no reference geometry of the site exists to score one against.

Back matter

1. *table row:* [8pt] **Date of Birth** | [to be completed]

2. *table row:* HIGH SCHOOL | [school, year — to be completed]

3. *table row:* **RESEARCH GRANT** | [to be completed, or "—"]